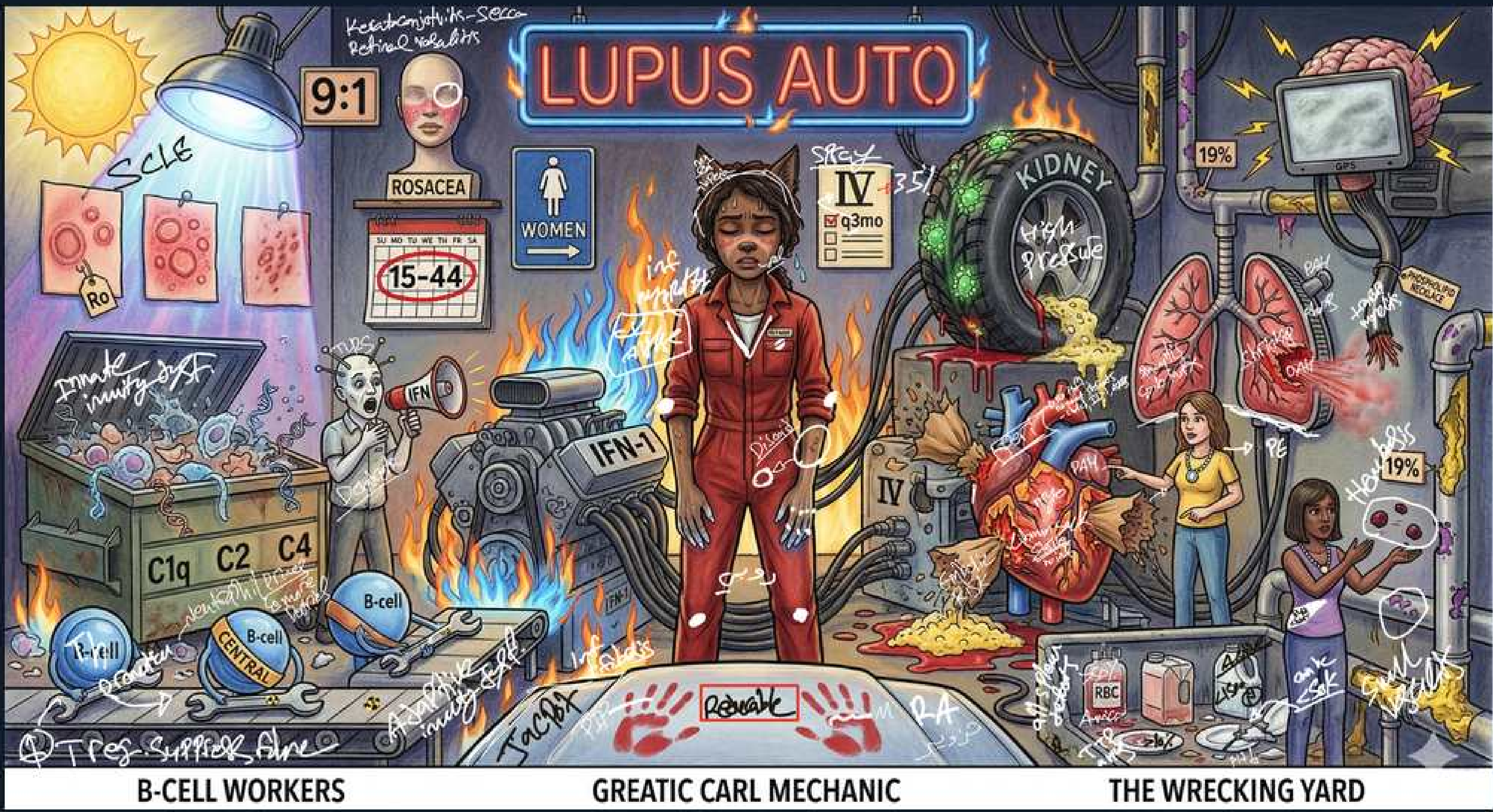
SLE — Overview, Demographics & Pathogenesis



1. LUPUS AUTO (autoimmune)

WHAT YOU SEE

Neon 'LUPUS AUTO' garage sign

WHAT IT ENCODES

SLE is a chronic multisystem autoimmune disease driven by loss of tolerance to self nuclear antigens, with tissue injury from type III (immune-complex) hypersensitivity plus type II antibody-mediated cytopenias. Immune complexes deposit in skin, kidney, joints, serosal surfaces, and vessels, activating complement and recruiting neutrophils. Classification (2019 EULAR/ACR) requires a positive ANA (titer ≥1:80) as the entry criterion, then weighted clinical/immunologic points ≥10. Mnemonic for criteria: 'SOAP BRAIN MD' (Serositis, Oral ulcers, Arthritis, Photosensitivity, Blood, Renal, ANA, Immunologic, Neuro, Malar, Discoid).

BOARD TRAP

A stem describing a single autoantibody attacking one organ (e.g. anti-TSH receptor → Graves, anti-acetylcholine receptor → myasthenia) is organ-specific autoimmunity, NOT SLE. SLE is defined by multisystem immune-complex injury with multiple antinuclear antibodies, so the discriminator is involvement of skin AND kidney AND joints AND blood, not one isolated target organ.

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2. 9:1 female predominance

WHAT YOU SEE

A clipboard reading 9:1 — the striking female-to-male ratio of SLE in the reproductive years.

WHAT IT ENCODES

SLE shows a striking ~9:1 female-to-male predominance during the reproductive years, reflecting estrogen's immunostimulatory effect and X-chromosome gene dosage (e.g. TLR7, escaping X-inactivation). Incidence is also 2–4x higher and more severe in Black, Hispanic, and Asian patients than in White patients. The female skew narrows toward ~2:1 in pre-pubertal and post-menopausal onset.

BOARD TRAP

Do NOT pick male-predominant or sex-equal. If a board stem features an older man with a sex ratio near 1:1 plus anti-histone antibodies after starting a drug, that is drug-induced lupus (no female predominance), not idiopathic SLE.

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3. Age 15-44, childbearing

WHAT YOU SEE

Calendar reading 15-44

WHAT IT ENCODES

Peak onset is in women of childbearing age, roughly 15–44 years, which is why pregnancy planning, contraception choice, and teratogen avoidance are core to management. Onset before puberty or after age 50 is atypical for classic SLE. New-onset 'lupus-like' disease in an older adult should prompt a search for a culprit drug.

BOARD TRAP

A 68-year-old man on hydralazine or procainamide who develops arthralgias, fever, and serositis with positive anti-histone but NEGATIVE anti-dsDNA and NORMAL complement has drug-induced lupus, not classic SLE. The age/sex profile and the absent renal/CNS disease are the discriminators.

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4. Sun / photosensitivity

WHAT YOU SEE

A bright sun and heat lamp blazing over the garage — UV light is the trigger that externalizes nuclear antigens and flares the disease.

WHAT IT ENCODES

UV light induces keratinocyte apoptosis, externalizing nuclear antigens (Ro/SSA) and driving the malar rash, subacute cutaneous lesions, and systemic flares. Photosensitivity is a classification criterion. Counsel strict photoprotection (broad-spectrum SPF 30+, avoid midday sun) and remember that hydroxychloroquine also has a photoprotective, anti-flare effect.

BOARD TRAP

A sun-exposed photosensitive rash with anti-Ro/SSA in a patient who is ANA-negative does not exclude lupus — subacute cutaneous LE and neonatal lupus are the classic anti-Ro syndromes that can be ANA-negative on some assays, so don't rule out lupus on a negative ANA alone in an anti-Ro stem.

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5. Subacute cutaneous LE (SCLE)

WHAT YOU SEE

'SCLE' rash plaques on wall, anti-Ro labeled

WHAT IT ENCODES

SCLE produces non-scarring annular/polycyclic or psoriasiform plaques on sun-exposed areas (upper back, shoulders, V of neck) and is strongly tied to anti-Ro/SSA. It is frequently DRUG-INDUCED (hydrochlorothiazide, terbinafine, calcium-channel blockers, TNF inhibitors, PPIs). Unlike discoid lupus it heals without scarring or atrophy.

BOARD TRAP

SCLE is anti-Ro/SSA-driven, NOT anti-dsDNA-driven. Choosing anti-dsDNA or low complement as the 'best associated antibody/finding' for SCLE is the trap — anti-dsDNA and hypocomplementemia track with lupus NEPHRITIS, not with the photosensitive SCLE rash.

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6. ANA face poster

WHAT YOU SEE

A face poster in the ROSACEA/ANA corner showing a butterfly malar rash — the cheek/nose rash that spares the nasolabial folds, screened by the sensitive-but-nonspecific ANA.

WHAT IT ENCODES

ANA is the sensitive (~95–99%) but non-specific screening test — a negative ANA makes SLE very unlikely, but a positive ANA occurs in many healthy people and other autoimmune diseases. The malar 'butterfly' rash is a fixed erythema over the cheeks and nasal bridge that characteristically SPARES the nasolabial folds. Confirmatory specific antibodies (anti-dsDNA, anti-Smith) follow a positive ANA.

BOARD TRAP

The classic discriminator: malar (butterfly) rash SPARES the nasolabial folds, whereas rosacea, seborrheic dermatitis, and the heliotrope/Gottron rash of dermatomyositis do NOT. A photodistributed rash that involves the nasolabial folds and has pustules points to rosacea, not SLE.

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7. Complement consumed: C1q C2 C4

WHAT YOU SEE

Bowling balls labeled C1q, C2, C4 tossed in a dumpster — complement being 'thrown away' (consumed) in active disease; deficiency of these also predisposes to lupus.

WHAT IT ENCODES

Active immune-complex disease CONSUMES complement, so low C3 and C4 (and high anti-dsDNA) signal an active flare, particularly lupus nephritis, and trend with disease activity. Separately, hereditary deficiency of early classical-pathway components (C1q > C4 > C2) is the strongest known genetic risk for SLE because impaired clearance of apoptotic debris and immune complexes breaks tolerance.

BOARD TRAP

Low C3/C4 means ACTIVE disease (consumption), not remission. The double trap: a patient with recurrent SLE and recurrent encapsulated/pyogenic or Neisseria-type infections may have an inherited early complement deficiency — so 'normal or high complement' in an active lupus flare is the wrong answer.

8. B-cell workers / autoantibodies

WHAT YOU SEE

Mechanic figures captioned B-CELL WORKERS churning out parts — the overactive B cells (driven by BAFF) mass-producing the autoantibodies and immune complexes.

WHAT IT ENCODES

B-cell hyperactivity, driven by BAFF/BLyS and T-cell help, produces the autoantibodies and immune complexes central to SLE. This rationalizes targeted biologics: belimumab (anti-BAFF) for active SLE and lupus nephritis, and rituximab (anti-CD20) and the newer anifrolumab (anti-type-I-interferon receptor) for refractory disease. Anti-dsDNA and anti-Smith are the most SPECIFIC antibodies; anti-Smith is the most specific overall.

BOARD TRAP

Anti-Smith is highly SPECIFIC but low sensitivity (~30%), so its absence does not exclude SLE. Confusing 'most sensitive' (ANA) with 'most specific' (anti-Smith / anti-dsDNA) is the classic antibody trap.

9. IFN-1 (type I interferon)

WHAT YOU SEE

IFN-1 stamped on the car's engine block — type I interferon (IFN-α) is the central engine driving lupus autoreactivity, the target of anifrolumab.

WHAT IT ENCODES

A type I interferon (IFN-α) gene signature is central to SLE pathogenesis: plasmacytoid dendritic cells sense nucleic-acid immune complexes via TLR7/9 and secrete IFN-α, amplifying autoreactivity. This pathway is the target of anifrolumab (anti-IFNAR1 monoclonal antibody). The same axis explains why some patients flare with viral infection.

BOARD TRAP

Type I interferon (IFN-α/β) drives SLE — do not confuse it with type II interferon (IFN-γ, the Th1/macrophage-activating cytokine of granulomatous disease). Picking IFN-γ or 'TNF-α as the central driver' is wrong; TNF-α inhibitors can even INDUCE a lupus-like syndrome.

10. Impaired apoptotic clearance

WHAT YOU SEE

A drunk, fumbling mechanic surrounded by uncleared debris — defective clearance of apoptotic nuclear debris leaves self-antigens exposed and breaks tolerance.

WHAT IT ENCODES

Defective clearance of apoptotic cell debris (from complement and macrophage dysfunction, plus excess NETosis) leaves nuclear antigens persistently exposed to the immune system, breaking self-tolerance and seeding immune complexes. This is why C1q/C4/C2 deficiency and neutrophil extracellular traps both promote lupus. It links the genetic complement story to the autoantibody output.

BOARD TRAP

The mechanism is impaired clearance of self/apoptotic debris (a tolerance defect), not a primary immunodeficiency that fails to clear PATHOGENS. A stem emphasizing recurrent severe infections as the core problem is describing immunodeficiency, not the apoptotic-clearance defect of SLE.

11. Lupus nephritis (kidney tire)

WHAT YOU SEE

A tire labeled KIDNEY caked with green immune-complex deposits — the 'full-house' deposition clogging the glomerulus, with class IV the worst.

WHAT IT ENCODES

Lupus nephritis is a leading cause of morbidity/mortality; subendothelial immune-complex deposition (a 'full-house' immunofluorescence pattern: IgG, IgA, IgM, C3, C1q) causes glomerulonephritis. Class IV (diffuse proliferative) is the most common and most severe, presenting with nephritic/nephrotic features, hypertension, and active urine sediment; biopsy guides therapy. Induction uses high-dose steroids plus mycophenolate mofetil OR low-dose cyclophosphamide (often with belimumab or voclosporin).

BOARD TRAP

Rising anti-dsDNA with FALLING C3/C4 signals active nephritis — picking 'normal complement, stable anti-dsDNA' as the marker of a flare is wrong. Also, you must BIOPSY to determine class before immunosuppression; treating empirically without histologic class is the management trap.

12. Serositis — lungs/pleura

WHAT YOU SEE

Inflamed, reddened lungs sitting in the wrecking yard — serositis of the pleura (pleuritis) giving a low-glucose exudative effusion.

WHAT IT ENCODES

Serositis (pleuritis and pericarditis) is a classification criterion and presents as pleuritic chest pain, exudative pleural effusions, or pericardial effusion. Lupus pleural fluid is an exudate with low glucose and may show low complement and LE cells. Pulmonary involvement also includes acute lupus pneumonitis and rare 'shrinking lung syndrome'.

BOARD TRAP

A pleural effusion in SLE is typically a low-glucose EXUDATE — calling it a transudate (as in heart failure) is wrong. Also distinguish lupus pleuritis from pulmonary embolism: in an SLE patient with pleuritic pain and clot, think antiphospholipid syndrome rather than simple serositis.

13. Cardiac / Libman-Sacks

WHAT YOU SEE

An inflamed red heart in the wrecking yard — pericarditis (commonest) plus sterile Libman-Sacks vegetations that can embolize.

WHAT IT ENCODES

Cardiac involvement spans pericarditis (most common), myocarditis, and Libman-Sacks endocarditis — sterile, verrucous, fibrin-platelet vegetations on BOTH surfaces of the mitral/aortic valves. These vegetations can embolize and are associated with antiphospholipid antibodies. Accelerated atherosclerosis from chronic inflammation makes coronary disease a leading long-term cause of death.

BOARD TRAP

Libman-Sacks vegetations are STERILE (non-bacterial thrombotic endocarditis), so blood cultures are negative and antibiotics are not the answer — the trap is treating it as infective endocarditis. Anticoagulation/treating the underlying APS, not antibiotics, is the management point.

14. Hematologic / thrombocytopenia

WHAT YOU SEE

Battered figures scattered in the wrecking yard — the destroyed blood cells: cytopenias (Coombs-positive AIHA, lymphopenia, thrombocytopenia).

WHAT IT ENCODES

Hematologic criteria include leukopenia (<4,000) or lymphopenia (<1,000), autoimmune hemolytic anemia (positive direct Coombs, high LDH, low haptoglobin, reticulocytosis), and thrombocytopenia (<100,000). Lymphopenia and a positive Coombs are especially characteristic. Anemia of chronic disease is the most common anemia overall, but Coombs-positive hemolysis is the immunologic hallmark.

BOARD TRAP

Thrombocytopenia in SLE can be from immune destruction (ITP-like) OR from antiphospholipid syndrome / thrombotic microangiopathy — the trap is assuming all SLE thrombocytopenia is benign ITP. A falling platelet count with thrombosis, schistocytes, or fetal loss points to APS, which is treated with anticoagulation, not just steroids.